DETECTION OF CIPHER TYPES USING ARTIFICIAL NEURAL NETWORKS

Abstract. The identification of a cryptosystem has been a challenge for decades. In this paper, the main objective is to identify the type of cryptosystem used to encrypt a certain piece of text. We have explored the realm of Artificial Neural Networks(ANNs) to recognize a pattern among complex classical cipher texts that generally have a simple representation in plain text. We have modelled our objective as a sequence-to-sequence learning task that we have tried to solve using Convolution Neural Networks (CNNs) and state-of-the-art Transformer models. With only a small dataset (130k) consisting of cipher texts and the corresponding cryptosystem used to encrypt the same, our model has shown a promising accuracy of 96.72 % which proves a significantly steep learning curve as compared to other sequence-to-sequence models. Here we show the enormous potential of these models and how they can perform even better if the barrier of resources and computation time is lifted.

Keywords: Artificial Neural Networks(ANNs)· Cryptosystem Identification · Classical Cipher text classification · Convolution Neural Networks(CNNs) · Transformer models.

1 Introduction

In today's world, information is a powerful tool. However, it is essential to protect information and data from unauthorized access, distortion, or any other unauthorized activity. Cryptography is the study of secret writing and is used to prevent unauthorized access to data. It works by using a key(s) to convert plaintext messages into unreadable ciphertext and vice versa. Encryption is the process of converting plaintext messages into an unreadable format to protect sensitive information (called ciphertext), while the inverse process is called decryption. Only authorized users can encrypt or decrypt messages [1].

The goal of the work is to use artificial neural networks to examine encrypted text files and classify them based on their cipher types. The process involves creating a collection of cipher texts with various cipher types, building a Deep Neural Network (DNN), and training it with a variety of cipher texts and training parameters to enable accurate learning and cipher text classification [2]. Once the DNN is trained, it can be used to determine the type of cipher text. In this work, the Caesar, Affine, Vigenre, Substitution, and Rail Fence ciphers were used to encrypt English texts. The encrypted files were then submitted to CNN and Transformer models, which made predictions, and the accuracy of these predictions was assessed.

Previous studies on the detection of the type of ciphers have used ANN, transformer, ensemble model of feature-engineering, and BLSTM-GRU cell unit network. The research which explored the use of Long Short-Term Memory (LSTM) and Transformer algorithms as feature-learning algorithms for the task of identifying 55 classical cipher types reported that with an ensemble model combining feature-engineering and feature-learning neural network types, they achieved an accuracy of 82.78% over a dataset of 10 million ciphertexts. One more work presented the progress of a research project that uses a TensorFlow-based network for detecting the cipher type. The network in this case was able to classify about 90% of the ciphertexts in a publicly available challenge correctly, and that their method resulted in a saving of about 54% computation time compared to using different solvers. Another study based on deep Bidirectional LSTM-Gated Recurrent Units (GRU) for ciphertext classification evaluated the model on two publicly available datasets and reported a high classification accuracy of 95.8%. In contrast, our study uses the newer version of transformers mainly the GPT and OPT models, to do the detection. Our results show that the use of this approach provides more accurate and reliable results compared to the previous ones. Our approach shows an improvement of 14% compared to them ensemble model of feature-engineering [3], 6% improvement compared to ANN [2] and 1.5% improvement compared to the BLSTM-GRU network model [4]. It is important to note that our results not only show the potential of these models but also highlight the possibility of even better performance if the limitations of resources and computation time are addressed.

This paper is organized into five different sections. Section 2 describes the preliminaries for this work followed by this introductory section. Section 3 elaborates on our proposed work with dataset generation and a discussion of the models. Section 4 shows the experimental findings. Section 5 contains the conclusion and the work that we plan to address in the future.

2 Preliminaries

This paper requires basic knowledge of neural networks and deep learning, as well as an overview of the encryption algorithms used. These algorithms can be useful for feature extraction. The main machine-learning topics covered are artificial neural networks (ANN), perceptrons, recurrent neural networks (RNN), convolutional neural networks (CNN), and transformers.

- **Artificial Neural Network (ANN)** An artificial neural network (ANN) is a type of computational model that mimics the human brain structurally and functionally [5]. ANNs consist of layers of interconnected neurons, which process input data and generate output values. ANNs are trained to perform numerous types of tasks, such as classification, regression, and clustering.
- **Perceptron** A perceptron is a type of artificial neural network (ANN) that can be used for binary classification tasks [6]. It consists of a single layer of neurons, with each neuron receiving input from multiple sources,

and producing a single output value. A set of weights determines the output of a perceptron. Those weights are then applied to the input values, and a threshold value is used to determine whether the output is 0 or 1.

- **Convolutional Neural Network (CNN)** CNNs consist of multiple layers of interconnected neurons, which process the input data in a hierarchical manner [7]. The first layer of a CNN typically consists of a set of filters. These filters are used extract features such as edges and textures by applying them to the input data. These features are then passed to the next layer of the CNN, where they are combined and processed to extract more complex features, such as shapes and objects. The output of a CNN is a set of probabilities that indicate the likelihood that the input data belongs to each of the possible classes.
- **Transformers** The transformer is a type of encoder-decoder model, which is commonly used for tasks such as machine translation and text summarization [8]. The use of self-attention, a mechanism that enables the model to learn relationships between different words in the input text, is the transformer's key innovation. This allows the transformer to process the input sentence in parallel, rather than in a sequential manner, which makes it much more efficient and scalable than previous NLP models.
- **Attention Mechanism** Self-attention is a the mechanism used in the transformer [8].With self-attention the model learns dependencies between different words in the input, by computing weighted sum of the input words based on their relevance to each other.

3 Proposed Work

3.1 Dataset Generation

The dataset has been generated for five types of ciphers:

- Caesar
- Affine
- Vigenre
- Substitution
- Rail Fence

The plaintext used for encryption was the texts from Shakespeares Hamlet [9], Macbeth [10], Merchant of Venice [11] and Romeo and Juliet [12].

We converted the text to lowercase and removed all whitespaces and punctuation. Then the plaintext was encrypted and exported as a CSV.

3.2 CNN Based Proposed Model

The main intuition behind this method is that CNN uses its kernel as a local receptive field [7], which means that a CNN kernel (small in size as we are dealing

with stream cipher and not block cipher) can look its nearby characters and will find some pattern in the ciphertext.

So, we first converted every ciphertext into their respective list of tokens using a TextVectorizer [13], and then we converted that list into an image as shown in Fig. 1.

$$\begin{bmatrix} a & b & c & d \end{bmatrix} \longrightarrow \begin{bmatrix} a & b & c & d \\ b & c & d & a \\ c & d & a & b \\ d & a & b & c \end{bmatrix}$$

Fig. 1: Converting text to 2D text matrix

Note that this conversion depends on the length of the string used, i.e. for a string for length n, we will generate a $n \times n$ matrix. After we generate all variable-sized matrices, we resize them to 64×64 using Bilinear Interpolation.

Now we can see that a kernel (say 3×3 with stride one) can observe its previous as well as its next two characters and can use that information to find patterns in the ciphertext.

We use a CNN to extract features from the matrix and then we feed those features to a Dense Network to classify between the ciphertext types [7].

3.3 Motivation Behind Choosing Transformer Model

In our case, we used pre-trained transformer models, which have been fine-tuned using our own dataset. This allowed us to leverage the powerful language understanding capabilities of the transformer architecture while also ensuring that the model was specifically tailored to our problem.

The transformer architecture is based on the idea of self-attention, which allows the model to weigh the importance of different tokens in the input sequence when making predictions [8]. This is done by computing a dot product between the input tokens and a set of learnable parameters, which are then used to generate a weight for each token. These weights are then used to compute a weighted average of the input tokens, which is used as the input to the next layer of the model.

This self-attention mechanism allows the transformer to effectively capture long-range dependencies between tokens, which is crucial for understanding the relationship between the ciphertext and the algorithm used to generate it.

In summary, we have used the transformer model as it is a state-of-the-art model for sequence-to-sequence problems and its self-attention mechanism and multi-head attention allow the model to capture long-range dependencies between tokens and attend to different parts of the input sequence in parallel, which is crucial for understanding the relationship between the cipher-text and the algorithm used to generate it.

3.4 Transformer Based Proposed Model

The Transformer in NLP is a novel architecture that aims to solve sequence-to-sequence tasks while effeciently handling long-range dependencies. The Transformer was proposed in the paper Attention Is All You Need [8]. The main intuition behind this method is to compute the relative importance of a ciphertext character with respect to the other ciphertext characters.

Our dataset contains encrypted text and the algorithm used for encryption, we create a text prompt from each row for example:

Cipher text: `ymccaygimgamkyyiycuycqocyeagmigkycmyyoyy`
Algorithm: `affine`

The model learns the relative importance of each token (character) and uses it to predict the type of ciphertext. For instance, if the model is trained using the previous prompt and we then provide the following prompt:

Cipher text: `yycmdkreqluithnalekstndonwothvn`
Algorithm:

The model may generate the output as affine which it learned from training.

In our use case, we have used the GPT-Neo (Generative Pre-Trained) [14] and OPT [15] models both with 125 million parameters to achieve our objective. GPT-Neo has been pre-trained on the Pile [16] dataset created by EleutherAI, whereas the OPT was trained on the Pile [16] as well as The Pushshift Reddit Dataset [17]. The GPT-Neo architecture is an extension to GPT2 [18] using local attention in every other layer with a window size of 256 tokens. On the other hand, the OPT's design is based on the latest GPT-3 [19] and both OPT and GPT-3 have a comparale performance. All these models are available for free in the Hugging Face [20] transformers library which can be used for a number of NLP tasks.

The following steps were performed:

1. Each data is converted into text prompts that is passed through a tokenizer that creates embedding for each token in our input string.
2. The input dataset has been split into the ratio 80:10:10 for training, testing, and validating purposes.
3. The training and validation set has been used to fine-tune the models hyperparameters specific to our use-case i.e. detecting the algorithm used in the ciphertext generation.
4. We used the fine-tuned model on our test dataset to check the accuracy of our model.
5. After fine-tuning, for each of our test data we performed sampling with the temperature set to 0.7 to return 10 predictions. The maximum predicted algorithm is chosen to be the correct one and is then compared with the actual value to compute the accuracy.

For our testing purpose, we used various decoding strategies to find the one which gives the best result. Out of the strategies: Greedy Search, Beam Search and Sampling with set temperature, we found sampling to give the most accurate results of all three.

The greedy method selects the next token that has the highest probability as the output while sampling selects one of the possible next tokens randomly. When we set a temperature, the randomness is lowered such that the likelihood of high-probability words is increased and that of the low probability is decreased. We found that a temperature set to 0.7 gave the optimum accuracy on our test data. 10 predictions are computed for each test data and the maximum predicted algorithm is chosen to be the correct one and is then compared with the actual value to compute the accuracy.

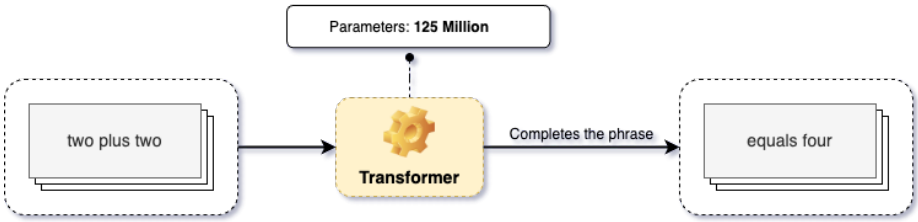


Fig. 2: Pre-training of a transformer model on a large corpus

Fig. 2 depicts the pre-training phase of a Transformer-based architecture. Upon completion of this phase, the model has acquired the capability of generating generic text. As an illustration, it can predict the continuation of the phrase ẗwo plus twoäs ëquals four.

Fig. 3 represents the pre-trained Transformer-based model that has undergone fine-tuning utilizing a custom dataset derived from several works of Shakespeare. The fine-tuned model is now capable of being applied to the specific task of identifying the cipher type of a given ciphertext.

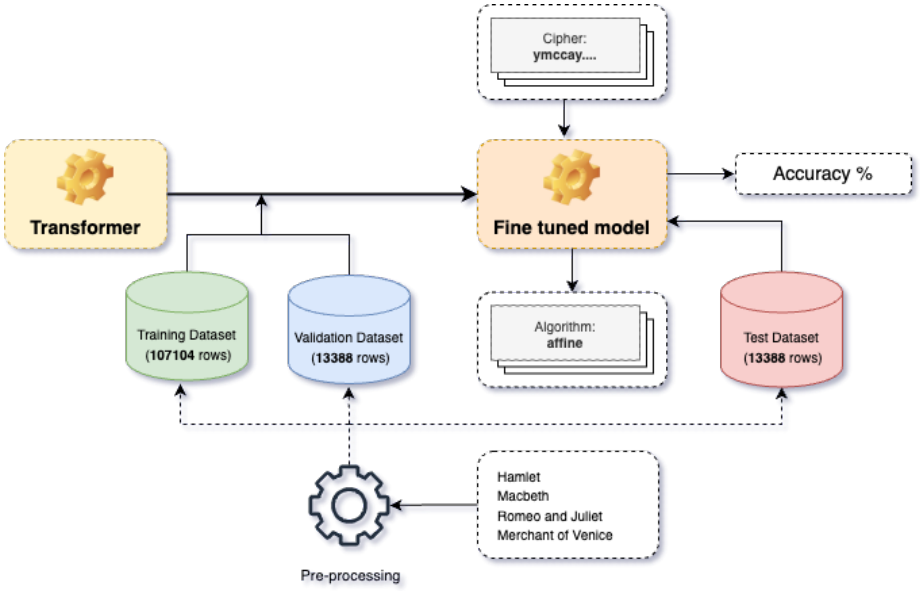


Fig. 3: Fine-tuning of the pre-trained transformer model on a custom dataset

4 Experimental Results

4.1 Dataset-1

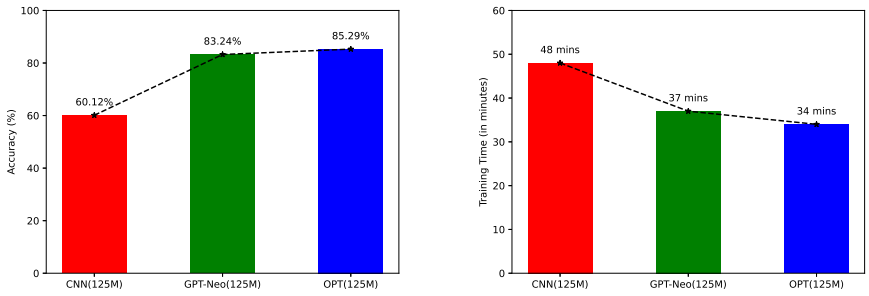
Our initial dataset consisted of 34075 rows that were split for training, testing, and validating in the ratio 80:10:10. The following are the sizes for each split:

- Train dataset: 27260 rows
- Validation dataset: 3408 rows
- Test dataset: 3407 rows

Table 1 shows the results obtained using the dataset-1 where GPT and OPT models are giving much better accuracy than CNN.

Table 1: Training time and accuracy on Dataset-1

Model	Number of Classes	Training time	Accuracy
CNN(125M)	5	48 mins	60.12%
GPT-Neo(125M)	5	37 mins	83.24%
OPT(125M)	5	34 mins	85.29%



(a) Accuracy (b) Training Time

Fig. 4: Comparison of Accuracies and Training Times on Dataset 1

4.2 Dataset-2

Next, we increased our dataset containing 133880 rows that were split for training, testing, and validating in the ratio 80:10:10. The following are the sizes for each split:

- Train dataset: 107104 rows
- Validation dataset: 13388 rows
- Test dataset: 13388 rows

Table 2 shows the results obtained using dataset-2:

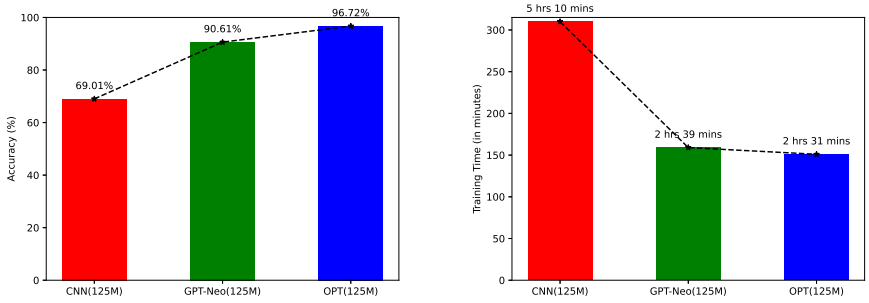
Table 2: Training time and accuracy on Dataset-2

Model	**Number of Classes**	**Training time**	**Accuracy**
CNN(125M)	5	5 hrs 10 mins	69.01%
GPT-Neo(125M)	5	2 hrs 39 mins	90.61%
OPT(125M)	5	2 hrs 31 mins	96.72%

Fig. 5 shows that the results obtained using OPT and GPT models are far better than other models based on CNN.

5 Conclusion

This paper illustrates the current progress of our work in the detection of ciphertext using Artificial Neural Networks. Here we used three types of Neural Networks to achieve our objective, namely: Convolution Neural Networks, GPT-Neo, and OPT.



(a) Accuracy (b) Training Time

Fig. 5: Comparison of Accuracies and Training Times on Dataset 2

With the CNN approach, our model was able to achieve an accuracy of 68.30% for 5 class classifications. We saw a significant improvement with the transformers approach using GPT-Neo on the same dataset and classes giving an accuracy of 90.61%. This was primarily due to the ability of GPT-Neo to memorize the local importance of tokens due to the self-attention feature. OPT gave better improvements, giving an accuracy of 96.72%, signifying that OPT is better than GPT-Neo for the task of detecting the type of ciphertext. We also observed that transformer-based models are giving better results than CNN while taking less time for training.

Our work can further be extended using similar transformer-based model like BERT, GPTJ, and Bloom to see if those models are better suited for the task. We will also increase the number of classes of ciphertext and the size of the dataset to see if we observe any improvements.

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